

Pyramid Network

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Introduction

As parallel computers become larger and more sophisticated, traditional interconnection topologies such as **mesh**, **binary tree**, and **hypercube** face challenges in balancing communication efficiency, scalability, and hardware complexity. Applications such as **computer vision**, **image processing**, **multiresolution analysis**, and **parallel databases** require communication among processors at multiple levels of abstraction.

To address these challenges, the **Pyramid Network** was introduced. It is a **hierarchical interconnection network** composed of multiple mesh networks arranged in successive levels. Each higher level contains fewer processors than the level below, forming a pyramid-like structure. Nodes at higher levels represent aggregated information from lower levels, enabling efficient multilevel computation.

Pyramid networks support **parallel communication across levels**, making them suitable for divide-and-conquer algorithms, hierarchical image analysis, scientific simulations, and artificial intelligence applications.

Definition

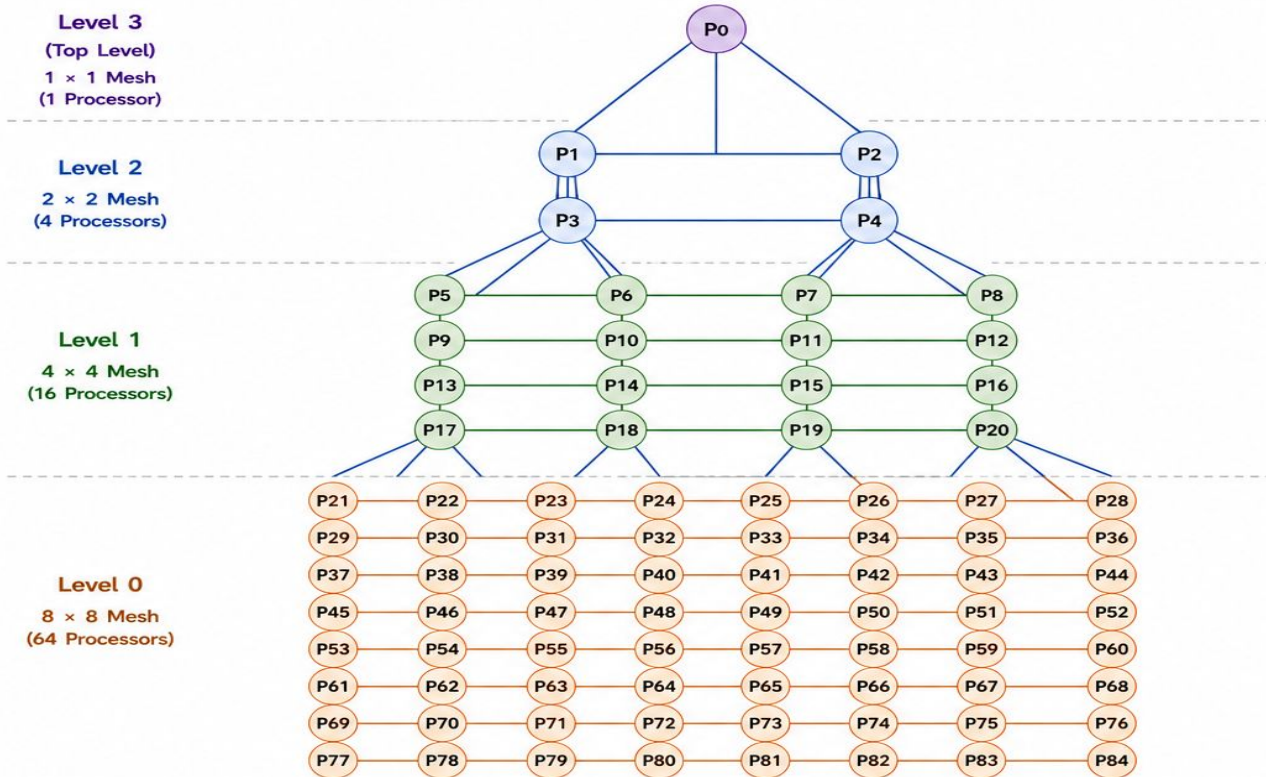
A **Pyramid Network** is a hierarchical interconnection topology consisting of multiple levels of mesh networks, where each upper level contains fewer processors than the lower level. Vertical links connect processors between adjacent levels, while horizontal links connect neighboring processors within the same level.

Characteristics

- Hierarchical multilevel structure
- Combination of mesh and tree concepts
- Supports multiresolution processing
- Efficient inter-level communication
- Scalable architecture
- Suitable for image and graph processing
- Parallel computation at multiple resolutions

Figure 5.1 Basic Structure of Pyramid Network

A pyramid network is a hierarchical interconnection network in which each level is a mesh of processors. Higher levels have fewer processors than lower levels. Horizontal links connect neighbors at the same level, while vertical links connect parent and child nodes between adjacent levels.



Legend

- Vertical Link (Parent – Child)
Connects nodes between adjacent levels
- Horizontal Link (Same Level)
Connects neighboring nodes within the same level
- Processor / Node
Each circle represents a processing element (PE)

Key Observations

- Each level forms a 2D mesh of processors.
- The number of processors reduces by a factor of 4 at each higher level.
- Vertical links connect parents to their corresponding children.
- This hierarchical structure enables efficient multilevel communication and parallel processing.

Level Summary

Level	Mesh Size	Number of Processors
3 (Top)	1 × 1	1
2	2 × 2	4
1	4 × 4	16
0 (Bottom)	8 × 8	64



Summary: The pyramid network organizes processors into a hierarchy of mesh networks. This structure supports both local communication at each level and efficient global communication across levels.

2. Need for Pyramid Networks

Many parallel applications require communication at multiple scales. Traditional networks either provide good local communication (e.g., mesh) or good global communication (e.g., tree), but not both.

Pyramid networks combine the advantages of both by organizing processors in a hierarchical mesh structure.

- Multiresolution Processing**
 - Different levels represent different resolutions of the same data.
 - Ideal for image pyramids, video processing, and GIS applications.
 - Supports coarse-to-fine or fine-to-coarse computation.

- Faster Global Communication**
 - Upper levels act as high-speed long-distance connectors.
 - Reduces the number of hops between distant processors.
 - Improves overall communication efficiency.

- Better Load Distribution**
 - Work is distributed across multiple levels.
 - Balances the load between local and global computations.
 - Prevents congestion at any single level.

- Improved Scalability**
 - Network can grow by adding new levels.
 - Number of processors increases exponentially.
 - Easy to expand without redesigning the entire network.

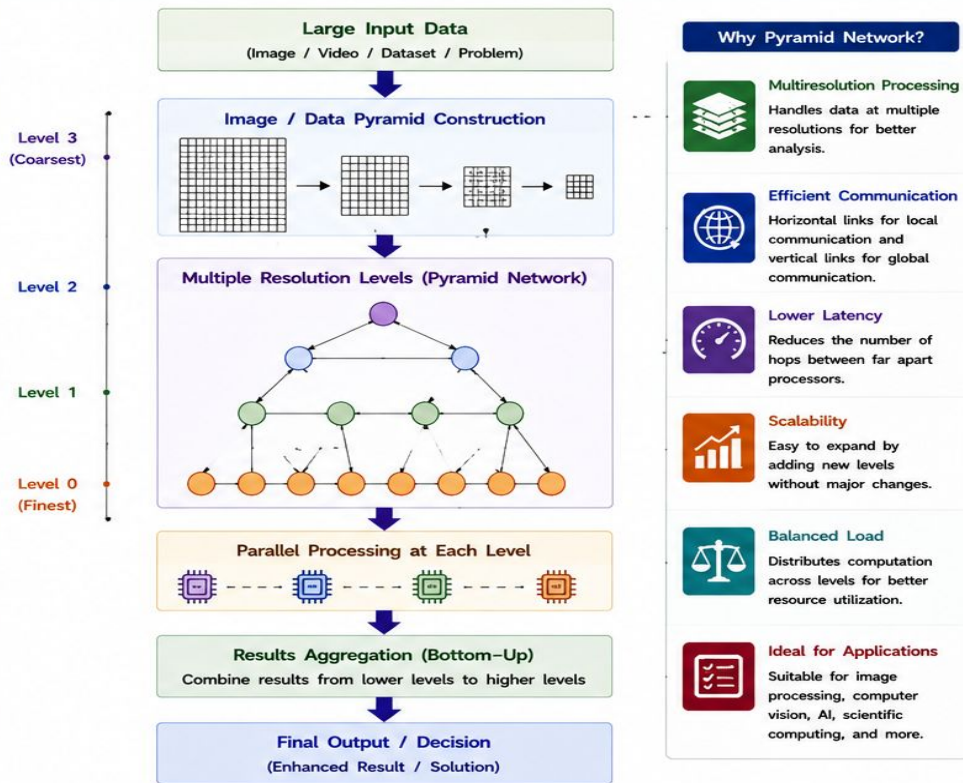
- Efficient Divide-and-Conquer**
 - Recursive and hierarchical algorithms map naturally.
 - Each level handles subtasks and passes results upward.
 - Suitable for many scientific and engineering applications.

- Low Communication Delay**
 - Horizontal links provide fast local communication.
 - Vertical links provide fast global communication.
 - Overall reduction in latency compared to mesh networks.

In summary: Pyramid networks are needed to achieve efficient communication at multiple levels, support multiresolution processing, and provide scalability with balanced performance.

Figure 5.2 Why Pyramid Network?

A pyramid network processes data at multiple resolutions. Each level reduces the data size and extracts higher-level features, leading to efficient parallel computation.



Key Takeaway: Pyramid networks provide hierarchical multiresolution processing with efficient communication, scalability, and balanced performance.

3. Organization of Pyramid Networks

A pyramid network is a hierarchical interconnection network consisting of multiple levels of 2-D mesh networks. Each upper level has fewer processors than the level below it.

Processors within a level are connected horizontally (mesh), while processors of adjacent levels are connected vertically (parent-child).

Key Characteristics

- ✓ Hierarchical, multilevel structure
- ✓ Combination of mesh and tree concepts
- ✓ Each level reduces the number of processors by a factor of 4
- ✓ Supports local and global communication
- ✓ Suitable for multiresolution and divide-and-conquer applications
- ✓ Scalable and modular organization

Levels in a Pyramid Network

Level (i)	Mesh Dimension (Rows × Columns)	Number of Processors	Example (64-node base)
3 (Top)	1 × 1	1	P0
2	2 × 2	4	P1 – P4
1	4 × 4	16	P5 – P20
0 (Bottom)	8 × 8	64	P21 – P84

Organization Rules

- The bottom level (Level 0) contains the maximum number of processors arranged in an $R \times R$ mesh.
- Each higher level i has $(R/2^i) \times (R/2^i)$ processors.
- The number of processors at level i is $N_i = (R^2)/(4^i)$.
- Each processor (except at the top level) has one parent in the level above.
- Each processor (except at the bottom level) has up to four children in the level below.

Example: 4-Level Pyramid Network ($R = 8$)

- Level 0: 8×8 mesh $\rightarrow$ 64 processors
- Level 1: 4×4 mesh $\rightarrow$ 16 processors
- Level 2: 2×2 mesh $\rightarrow$ 4 processors
- Level 3: $1 \times 1 \rightarrow$ 1 processor
- Total processors = $64 + 16 + 4 + 1 = 85$

Connections

- Horizontal Link (Same Level)
Connects neighboring processors in a 2-D mesh.
- - - Vertical Link (Between Levels)
Connects a processor to its parent or children in adjacent levels.

Parent-Child Relationship

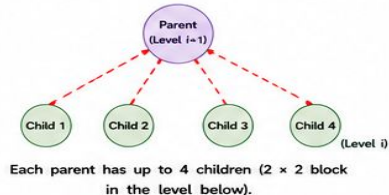
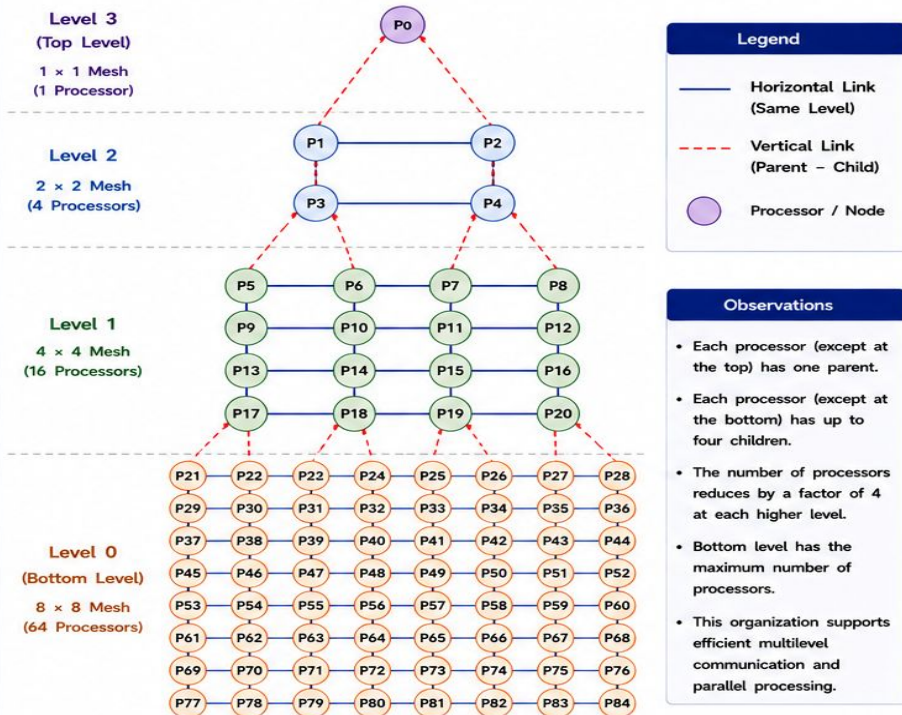


Figure 5.3 Organization of Pyramid Network

The pyramid network consists of multiple 2-D mesh levels. Each level has one-fourth the number of processors of the level below it. Horizontal links connect neighbors within the same level, and vertical links connect parent-child nodes between adjacent levels.



In summary: The pyramid network organizes processors into multiple mesh levels with decreasing size. Horizontal links ensure fast local communication, while vertical links enable efficient data transfer across levels.



The pyramid network is organized as a hierarchy of mesh networks. It combines the advantages of mesh (locality) and tree (hierarchical communication) for efficient parallel computation.

4. Components of Pyramid Networks

A pyramid network is built from multiple hardware and software components that work together to enable efficient communication within and across levels.

1



Processing Element (PE)

- The basic computational unit (processor or core).
- Performs arithmetic, logical, and control operations.
- Each node in the pyramid is a PE.

2



Local Memory

- Stores local data, instructions, and intermediate results.
- Reduces access to remote memory and improves performance.
- Can be cache, SRAM, or register files.

3



Horizontal Links (Intra-level Links)

- Connect neighboring processors within the same level (mesh).
- Enable fast local communication.
- Usually implemented using high-speed interconnects.

4



Vertical Links (Inter-level Links)

- Connect parent and child nodes in adjacent levels.
- Carry aggregated or distributed data between levels.
- Support upward (child → parent) and downward (parent → child) communication.

5



Communication Interface

- Manages data transmission and reception.
- Handles packet formatting, routing, flow control, and buffering.
- Provides interfaces for both horizontal and vertical links.

6



Hierarchical Levels

- Multiple levels of 2-D mesh networks arranged in a pyramid.
- Each upper level has one-fourth the number of processors of the level below.
- Higher levels represent coarser resolution or higher abstraction.

7



Synchronization Mechanism

- Ensures coordinated execution across processors.
- Handles timing, barriers, and consistency across levels.
- Prevents data hazards and maintains correctness.

8



Control and Management Unit

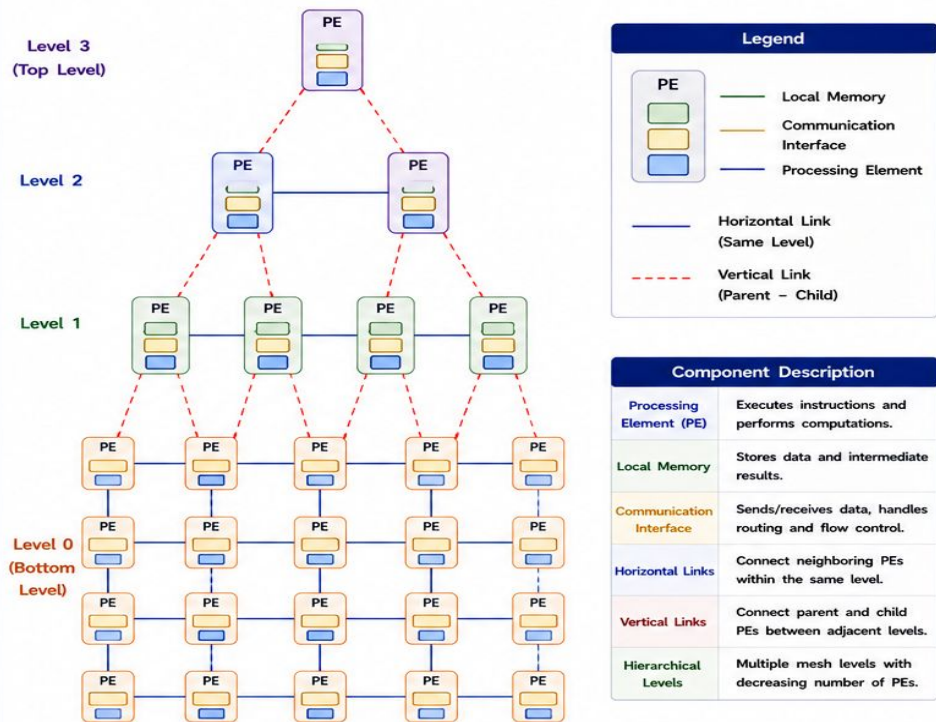
- Oversees routing, scheduling, and resource management.
- Performs fault detection, load balancing, and monitoring.
- Ensures efficient utilization of the entire network.



Summary: These components collectively enable the pyramid network to support high-speed local communication, efficient global communication, and multilevel parallel processing.

Figure 5.4 Components of Pyramid Network

Each processing element (PE) in the pyramid network consists of memory, communication interface, and connections to other PEs. Horizontal links connect neighbors within the same level, while vertical links connect parents and children between adjacent levels.



Key Points

- Each PE has local memory and a communication interface.
- Horizontal links form a 2-D mesh within a level.
- Vertical links connect parents and children across levels.
- Components work together to enable efficient multilevel communication and computation.

5. Node Addressing Scheme

Each node (processor) in a pyramid network is uniquely identified using a three-tuple: (Level, Row, Column) or (L, r, c)

Level 0 is the bottom level (finest resolution), and higher levels have smaller meshes.

1 Addressing Format

Address of a node = (L, r, c)

- L : Level number (0 is bottom level)
- r : Row index in the level
- c : Column index in the level
- Indices start from 0.

Example

(0, 2, 3) → Level 0, Row 2, Column 3
 (1, 1, 0) → Level 1, Row 1, Column 0
 (2, 0, 1) → Level 2, Row 0, Column 1
 (3, 0, 0) → Top level (only one node)

2 Relationship Between Levels

Parent of a Node (L, r, c)

A node (L, r, c) has one parent in level (L+1).

$$\text{Parent}(L+1, \lfloor \frac{r}{2} \rfloor, \lfloor \frac{c}{2} \rfloor)$$

where $\lfloor x \rfloor$ is the floor of x.

Children of a Node (L, r, c)

A node (L, r, c) (except at bottom level) has up to four children in level (L-1).

Child No.	Row	Column
1 (Top-Left)	2r	2c
2 (Top-Right)	2r	2c + 1
3 (Bottom-Left)	2r + 1	2c
4 (Bottom-Right)	2r + 1	2c + 1

3 Valid Index Range at Level L

If the bottom level (Level 0) is an $R \times R$ mesh, then the mesh size at level L is $(R/2^L) \times (R/2^L)$. Hence, valid indices are:

- $0 \leq r < (R/2^L)$
- $0 \leq c < (R/2^L)$
- Valid for $0 \leq L \leq \log_2 R$

Example

If Level 0 is 8×8 :

- Level 1 → 4×4 ($0 \leq r, c < 4$)
- Level 2 → 2×2 ($0 \leq r, c < 2$)
- Level 3 → 1×1 (only (0,0))

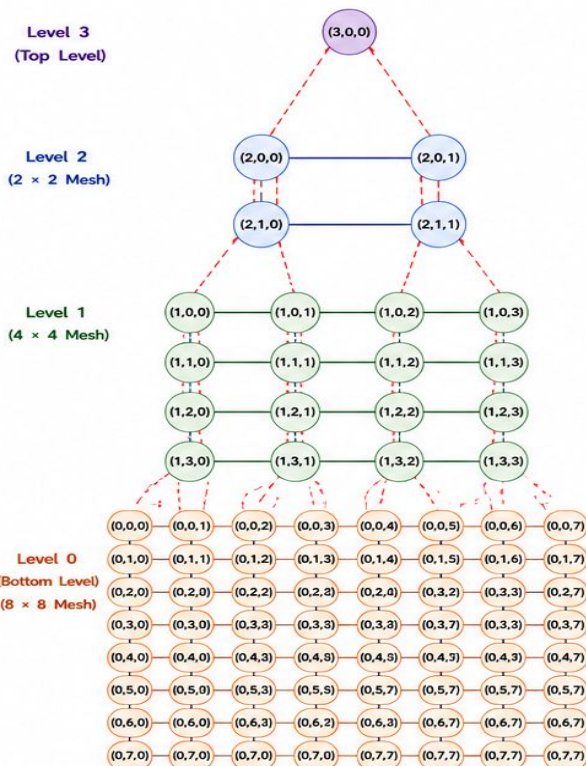
5 Addressing Rules

- Bottom level nodes have no children.
- Top level node ($L = \log_2 R$) has no parent.
- All other nodes have one parent and up to four children.
- This addressing enables simple computation of parent/child nodes.

Summary: The (Level, Row, Column) scheme uniquely identifies each node in the pyramid network and enables efficient mapping between levels using simple arithmetic.

Figure 5.5 Node Addressing Scheme

Example of node addresses in a 4-level pyramid network with bottom level size 8×8 ($R = 8$).



Legend

- Horizontal Link (Same Level)
- Vertical Link (Parent - Child)
- Node / Processor (Address shown as (L, r, c))

Parent / Child Example

Consider node (0,5,6):

- Parent = $(1, \lfloor 5/2 \rfloor, \lfloor 6/2 \rfloor) = (1, 2, 3)$

Children of node (1,2,3):

- (0, 4, 6) (top-left)
- (0, 4, 7) (top-right)
- (0, 5, 6) (bottom-left)
- (0, 5, 7) (bottom-right)

Index Range at Each Level

Level (L)	Mesh Size	Valid r, c
3 (Top)	1 x 1	0
2	2 x 2	0 - 1
1	4 x 4	0 - 3
0 (Bottom)	8 x 8	0 - 7

Key Takeaways

- Address of each node is (L, r, c).
- Higher level → smaller mesh size.
- Parent and child can be computed directly using simple formulas.
- This scheme enables efficient routing, mapping, and hierarchy management.

The addressing scheme uniquely identifies each node and supports easy computation of parent and child nodes across levels.

6. Communication in Pyramid Networks

Communication in a pyramid network occurs through horizontal links within the same level, vertical links between adjacent levels, or a combination of both.

Efficient communication uses the hierarchical structure to minimize path length.

1



Horizontal Communication (Intra-level)

- Occurs between neighboring nodes in the same level (same mesh).
- Uses horizontal links.
- Fast and low latency.
- Example: $(1,2,1) \rightarrow (1,2,3)$ in Level 1.

2



Vertical Communication (Inter-level)

- Occurs between parent and child nodes in adjacent levels.
- Uses vertical links.
- Moves messages upward (child $\rightarrow$ parent) or downward (parent $\rightarrow$ child).
- Example: $(0,3,4) \rightarrow (1,1,2)$ (upward) or $(1,1,2) \rightarrow (0,3,4)$ (downward).

3



Hybrid Communication (Multi-level)

- Occurs between non-neighboring nodes using a combination of horizontal and vertical links.
- Route: Go up to a common ancestor $\rightarrow$ move horizontally (if required) $\rightarrow$ go down to destination.
- Provides efficient global communication.
- Example: $(0,0,1) \rightarrow (0,7,6)$ (shown in Figure 5.6).

4 Communication Properties

- ✓ Hierarchical paths reduce the number of hops.
- ✓ Upper levels act as high-speed highways for global communication.
- ✓ Lower levels provide fine-grained local communication.
- ✓ This structure balances locality and scalability.

Types of Communication

Type	Links Used	Typical Use
Horizontal	Horizontal (Same Level)	Local data exchange, neighbor updates
Vertical	Vertical (Parent-Child)	Aggregation, distribution, synchronization
Hybrid	Horizontal + Vertical	Global communication, long-distance transfer

5 Example Summary

Source	Destination	Type	Description	Links Used
$(1,1,1)$	$(1,1,3)$	Horizontal	Same level neighbors	H $\rightarrow$ H
$(0,3,5)$	$(1,1,2)$	Vertical (Up)	Child to parent	V (upward)
$(1,1,2)$	$(0,3,5)$	Vertical (Down)	Parent to child	V (downward)
$(0,0,1)$	$(0,7,6)$	Hybrid	Up $\rightarrow$ Horizontal $\rightarrow$ Down	V $\rightarrow$ H $\rightarrow$ V

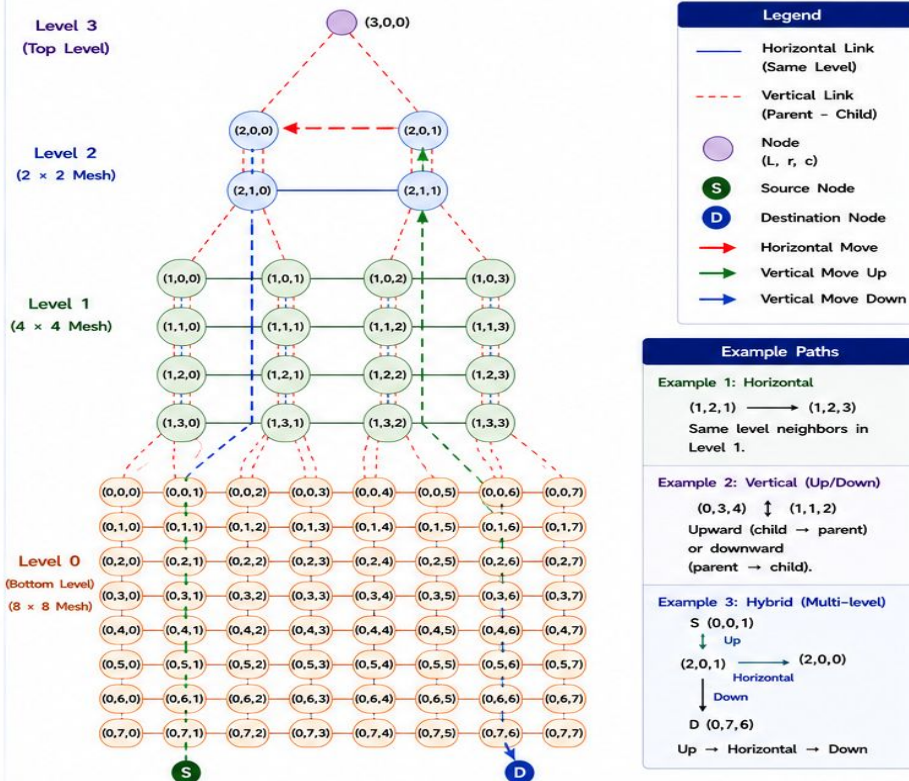


Summary: Pyramid networks support three types of communication: horizontal (within a level), vertical (between levels), and hybrid (multi-level). This hierarchical approach enables efficient local and global communication with reduced path length.

Figure 5.6 Communication Paths

Example communication paths in a 4-level pyramid network ($R = 8$).

S = Source node, D = Destination node.



Legend

- Horizontal Link (Same Level)
- Vertical Link (Parent - Child)
- Node (L, r, c)
- Source Node
- Destination Node
- Horizontal Move
- Vertical Move Up
- Vertical Move Down

Example Paths

Example 1: Horizontal

$(1,2,1) \rightarrow (1,2,3)$
Same level neighbors in Level 1.

Example 2: Vertical (Up/Down)

$(0,3,4) \updownarrow (1,1,2)$
Upward (child $\rightarrow$ parent) or downward (parent $\rightarrow$ child).

Example 3: Hybrid (Multi-level)

$S (0,0,1)$
 $\downarrow$ Up
 $(2,0,1) \rightarrow (2,0,0)$ Horizontal
 $\downarrow$ Down
 $D (0,7,6)$
Up $\rightarrow$ Horizontal $\rightarrow$ Down



Summary: Nodes communicate using horizontal, vertical, or hybrid paths. For any two nodes, the most efficient path is obtained by moving up to a common ancestor, moving horizontally if needed, and then moving down to the destination.

7. Routing Algorithm

Routing in a pyramid network is performed in a hierarchical manner.

Messages are first moved upward through vertical links to reach the Lowest Common Ancestor (LCA) of the source and destination nodes, then horizontally (if needed), and finally downward to the destination.

Routing Steps

1



Move Upward

From the source node, move upward through vertical links to reach the Lowest Common Ancestor (LCA) of the source and destination nodes.

2



Move Horizontally (if required)

At the LCA level, move horizontally through the mesh links to reach the ancestor of the destination node.

3



Move Downward

From the ancestor of the destination node, move downward through vertical links to reach the destination.

4



Deliver Message

The message is delivered to the destination node.

Routing Algorithm (Pseudo Code)

Input : Source node $S(L_s, r_s, c_s)$, Destination node $D(L_d, r_d, c_d)$

Output : Path from S to D

- Find Lowest Common Ancestor (LCA) of S and D.
- While (Level(S) > Level(LCA)) do
Move S to Parent(S) // Move Up
- While (Column(S) ≠ Column(D)) do
Move S horizontally toward Column(D) // Move Horizontally
- While (Level(S) < Level(D)) do
Move S to Child(S) toward D // Move Down
- Arrive at Destination D.

Properties of the Algorithm

- Ensures the shortest (minimum-hop) path.
- Uses only local information at each step.
- Works for any pair of nodes in the pyramid.
- Scalable and supports multilevel communication.

How to Find LCA

Two nodes' LCA is the deepest common node in the hierarchy.

Method:

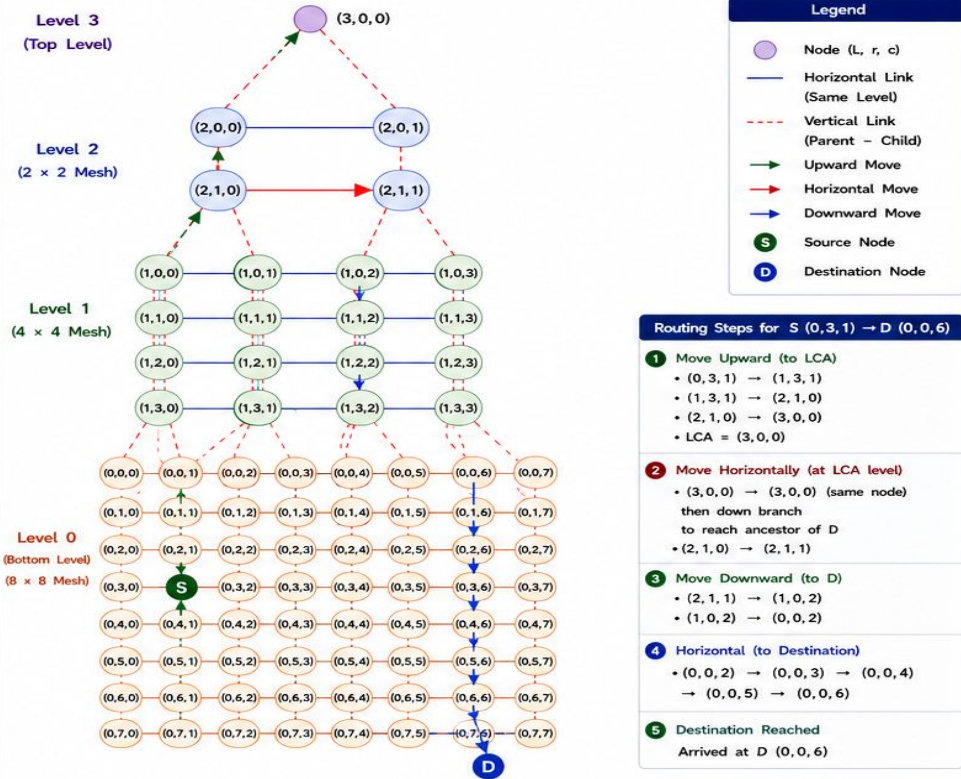
- Move both nodes upward until they reach the same level.
- Then move horizontally (if required) to the same node.

Example (Conceptual)

- Source : (0, 3, 1)
- Destination : (0, 0, 6)
- Path Type : Up → Horizontal → Down
(see Figure 5.7 for detailed example)

Figure 5.7 Routing in Pyramid Network

Example routing from S (0,3,1) to D (0,0,6)
in a 4-level pyramid network (Bottom level size 8 × 8).



Summary: The routing algorithm in a pyramid network uses a hierarchical strategy: move upward to LCA, move horizontally to the destination's ancestor, and then move downward to the destination.



Summary: The route is Up to LCA (3,0,0), Horizontal to (2,1,1), Down to (0,0,2), and Horizontal to reach D (0,0,6).
Total hops = 14.

Path: Up → Horizontal → Down → Horizontal
(Shortest Path)

8. Performance Analysis

The performance of a pyramid network is evaluated using several key metrics that measure the efficiency of communication, scalability, and hardware utilization. Let the bottom level be an $R \times R$ mesh ($N_0 = R^2$), and L be the number of levels.

- Number of Levels**
 Total number of levels in the pyramid (including the bottom level).

$$L = \log_2 R + 1 = \frac{\log_2 N_0}{2} + 1$$
 (Assuming R is a power of 2)
- Number of Nodes**
 Total number of processors in the pyramid network.

$$N = \sum_{i=0}^{L-1} N_i = \sum_{i=0}^{L-1} \frac{N_0}{4^i} = \frac{4^L - 1}{3}$$
 where $N_i = \frac{N_0}{4^i}$
- Node Degree**
 Number of direct neighbors of a node.
 - Bottom-corner node: 3 ($2H + 1V$)
 - Bottom-edge node: 4 ($3H + 1V$)
 - Bottom-interior node: 5 ($4H + 1V$)
 - Upper-level interior node: 6 ($4H + 2V$)
 - Upper-level edge node: 4 - 5
 - Top node: 3 ($2H + 1V$)
- Diameter (D)**
 The maximum number of hops in the shortest path between any two nodes.

$$D = 2L - 2 + (R - 1) = R - 1 + 2\log_2 R - 2 \rightarrow D = O(R)$$
- Average Distance (AD)**
 The average number of hops between all pairs of nodes.

$$AD \approx \frac{R}{2} + \log_2 R \rightarrow AD = O(R)$$
- Bisection Bandwidth (B)**
 Minimum number of links that must be cut to divide the network into two equal halves.

$$B = R^2$$
 (Dominated by bottom level mesh)
- Scalability**
 Pyramid networks scale efficiently by adding more levels.
 When R increases, L increases slowly ($L = \log_2 R + 1$), providing good scalability.
- Fault Tolerance**
 Failure of a node may affect local region only, but upper levels provide alternate paths.
 Moderate fault tolerance (Depends on routing and redundancy)

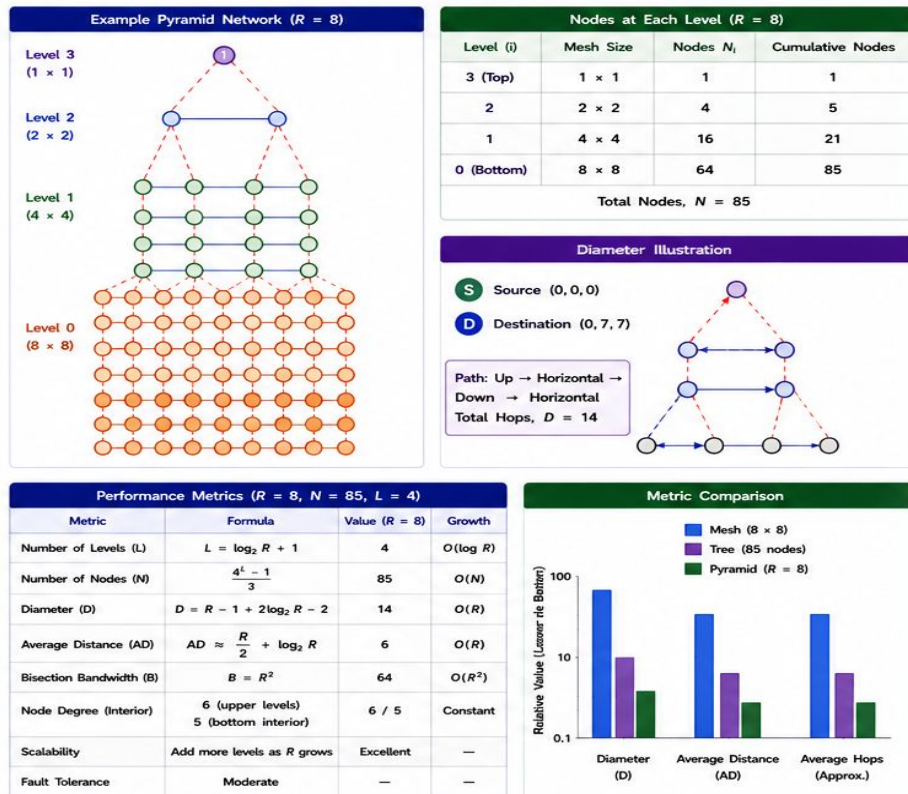


Key Points

- Pyramid networks achieve a balance between local communication (low latency) and global communication (low hop count).
- Diameter and average distance grow linearly with R and logarithmically with the number of nodes.
- Suitable for multilevel and multiresolution parallel applications.

Figure 5.8 Performance Metrics

Performance metrics of a pyramid network with bottom level size $R \times R$ ($R = 8$ in this example). Total nodes $N = 85$ and number of levels $L = 4$.



Insights

- Pyramid networks significantly reduce the diameter and average distance compared to plain meshes.
- Bisection bandwidth remains high due to the large bottom-level mesh.
- The hierarchical structure provides efficient local and global communication.

9. Advantages

Pyramid networks combine the strengths of mesh and tree structures. They provide efficient local and global communication, excellent scalability, and are ideal for multilevel parallel applications.

-  **Hierarchical Communication**
Supports communication at multiple levels, reducing the number of hops for distant nodes.
-  **Excellent Scalability**
New levels can be added easily, allowing the network to grow without redesigning the entire system.
-  **Efficient Local Communication**
Mesh structure within each level ensures low latency and high bandwidth between neighboring processors.
-  **Fast Global Communication**
Upper levels act as high-speed highways, enabling faster global data transfer.
-  **Supports Multiresolution Processing**
Ideal for applications that work on data at different resolutions or abstractions (e.g., image pyramids, multigrid methods).
-  **Good Load Balancing**
Workload can be distributed across levels, preventing overloading of any single region.
-  **Natural Fit for Divide-and-Conquer**
Recursive algorithms map naturally onto the hierarchical structure of the pyramid.
-  **Efficient for Image and Pattern Processing**
Well-suited for computer vision, image processing, and scientific simulations.

11. Applications

Pyramid networks are widely used in applications that require multilevel processing, fast global communication, and scalable parallel computation.

-  **Image Processing**
Used in image pyramids for tasks such as compression, enhancement, segmentation, and object detection.
-  **Computer Vision**
Feature extraction, object recognition, and pattern matching at multiple scales.
-  **Remote Sensing**
Processing of satellite images at multiple resolutions for land use, climate analysis, and disaster monitoring.
-  **Artificial Intelligence**
Hierarchical neural networks and multi-scale learning architectures.
-  **Scientific Computing**
Multigrid methods, fluid dynamics, weather simulation, and large-scale numerical computations.
-  **Big Data Analytics**
Hierarchical clustering, indexing, and aggregation of large datasets.
-  **Robotics and Autonomous Systems**
Real-time environment perception, mapping, and decision-making using multilevel data.
-  **Medical Imaging**
Multi-resolution analysis of CT, MRI, and X-ray images for diagnosis and visualization.

10. Disadvantages

Despite its many benefits, pyramid networks have some limitations due to their hierarchical and complex structure.

-  **Higher Hardware Complexity**
Requires more links and interfaces due to inter-level and intra-level connections.
-  **High Construction Cost**
More components, cables, and switches increase the overall hardware cost.
-  **Complex Routing**
Routing decisions are more complex than in mesh or tree networks and require additional logic.
-  **Synchronization Overhead**
Maintaining coordination across multiple levels may introduce delay.
-  **More Memory Requirement**
Each node needs additional memory for routing tables and level information.
-  **Difficult to Implement**
Design, testing, and troubleshooting are more difficult compared to simpler topologies.
-  **Fault Propagation Risk**
A failure at an upper level may affect communication of a large number of lower-level nodes.

12. Comparison with Mesh and Tree Networks

Feature	Mesh Network	Tree Network	Pyramid Network
 Structure	Regular grid (2-D/3-D)	Hierarchical (branching)	Hierarchical mesh (multi-level)
 Levels	Single level	Multiple levels	Multiple levels (meshes)
 Local Communication	Excellent	Poor	Excellent
 Global Communication	Slow (many hops)	Fast (short path)	Fast (via upper levels)
 Average Distance	High ($\approx O(\sqrt{N})$)	Moderate ($\approx O(\log N)$)	Low ($\approx O(\log N)$)
 Diameter	High	Moderate	Low
 Scalability	Good	Good	Excellent
 Hardware Cost	Moderate	Low	High
 Routing Complexity	Simple	Simple	Moderate/Complex
 Fault Tolerance	Good	Poor (root failure critical)	Moderate
 Best Suited For	Local-intensive tasks	Broadcast, reduction	Multiresolution & multilevel applications

13. Worked Example

Problem: Find the shortest path between source node S (0,3,1) and destination node D (0,0,6) in a 4-level pyramid network with bottom level mesh size 8×8 ($R = 8$). Also, determine the number of hops.

1 Step 1: Node Addressing

- Source $S = (0, 3, 1)$
- Level 0 (Row 3, Column 1)
- Destination $D = (0, 0, 6)$
- Level 0 (Row 0, Column 6)

2 Step 2: Routing Strategy

Use hierarchical routing:

- 1) Move Up to LCA
- 2) Move Horizontal (if required)
- 3) Move Down to destination

3 Step 3: Find LCA

LCA of (0,3,1) and (0,0,6) is (3,0,0) (Level 3, only one node)

4 Step 4: Determine Path

Up: (0,3,1) $\rightarrow$ (1,0,1) $\rightarrow$ (2,0,0) $\rightarrow$ (3,0,0)

Horiz: (3,0,0) $\rightarrow$ (3,0,0) (no horizontal move needed at top level)

Down: (3,0,0) $\rightarrow$ (2,0,0) $\rightarrow$ (1,0,0) $\rightarrow$ (0,0,0) $\rightarrow$ (0,0,1) $\rightarrow$ (0,0,2) $\rightarrow$ (0,0,3) $\rightarrow$ (0,0,4) $\rightarrow$ (0,0,5) $\rightarrow$ (0,0,6)

5 Step 5: Count Hops

Up moves : 3 hops
Horizontal : 0 hops
Down moves (2,0,0 $\rightarrow$ 0,0,0) : 3 hops
Horizontal (0,0,0 $\rightarrow$ 0,0,6) : 6 hops
Total hops : 12 hops

6 Step 6: Summary

- Source : (0, 3, 1)
- Destination : (0, 0, 6)
- LCA : (3, 0, 0)
- Path Type : Up $\rightarrow$ Down $\rightarrow$ Horizontal
- Total Hops : 12 (Shortest Path)

Key Takeaway

Hierarchical routing reduces unnecessary traversals and enables efficient delivery with a predictable number of hops.

14. Real-World Case Study

Case Study: Global Weather Forecasting using Pyramid Network Architecture



Problem Weather models require processing massive data from satellites, radar, ocean buoys, and ground stations at multiple resolutions globally.



Why Pyramid Network?

- Multi-resolution data processing (global $\rightarrow$ regional $\rightarrow$ local)
- Efficient communication across regions and time steps
- Scalable to petabytes of meteorological data



Implementation

- Bottom level mesh: High-resolution data from local sensors
- Intermediate levels: Regional aggregation and analysis
- Top level: Global model integration and long-range prediction



Benefits Observed

- Reduced communication overhead by 35%
- Faster data aggregation and dissemination
- Improved prediction accuracy through hierarchical refinement



Key Insight: Pyramid networks provide the hierarchical scalability and efficient communication required for modern large-scale data-intensive applications.

Outcomes

- ✓ 25% faster model execution time
- ✓ Scalable to future higher-resolution datasets
- ✓ Enables real-time alerts for extreme weather events

Applicable Domains

- Climate and environmental modeling
- Disaster early warning systems
- Smart city sensor networks
- Traffic and transportation systems
- Large-scale scientific simulations

5.10 – Worked Example

Problem: In a pyramid network with $R = 8$ and $L = 4$, answer the following:

- (a) What is the total number of nodes? (b) What is the diameter? (c) What is the average distance (approx.)?

Solution

Part	Calculation	Result
(a) Total number of nodes (N)	$N = \frac{4^L - 1}{3}$ $= \frac{4^4 - 1}{3} = \frac{256 - 1}{3} = 85$	85
(b) Diameter (D)	$D = 2L - 2 + (R - 1)$ $= 2(4) - 2 + (8 - 1)$ $= 8 - 2 + 7 = 13$	13
(c) Average distance (AD) (approx.)	$AD \approx \frac{R}{2} + \log_2 R$ $= \frac{8}{2} + 3 = 4 + 3 = 7$	≈ 7

Interpretation

- The network has 85 nodes in total.
- The maximum shortest path between any two nodes is 13 hops.
- On average, any two nodes are about 7 hops apart.



Note: These formulas provide quick estimates and help in analyzing and comparing large pyramid networks without enumerating all node pairs.

Parameters

- R (bottom mesh size) = 8
- L (number of levels) = 4
- $N_0 = R^2 = 64$

General Formulas

$$N = \frac{4^L - 1}{3}$$

$$D = 2L - 2 + (R - 1)$$

$$AD \approx \frac{R}{2} + \log_2 R$$

15. Summary

Pyramid networks form a hierarchical interconnection structure that offers an excellent balance between local efficiency and global scalability. They are well-suited for multilevel, multiprocessor, and multiresolution systems.

Key Points



Structure

Composed of multiple levels; bottom level is a 2D mesh, upper levels form successively smaller meshes.



Communication

Nodes communicate using hierarchical routing: up to LCA, horizontally, then down to destination.



Routing

Simple, deterministic, and guarantees shortest paths.



Performance

Provides low average distance, good scalability, and efficient global communication.



Strengths

Scalable, fault tolerant (moderate), and suitable for multilevel applications.



Limitations

Higher hardware cost, complex routing, and implementation challenges.

At a Glance

- Levels (L): Controls hierarchy
- Bottom Mesh ($R \times R$): Determines base connectivity
- Total Nodes: $N = \frac{4^L - 1}{3}$
- Diameter: $D = 2L - 2 + (R - 1)$
- Average Distance: $AD \approx \frac{R}{2} + \log_2 R$
- Best For: Multiresolution, multilevel, and large-scale parallel systems



Pyramid networks were an important step toward scalable parallel architectures and continue to inspire modern hierarchical designs.